

MODEL QUESTION PAPER
POLYMER TECHNOLOGY

Time: 3 hours

(Maximum marks : 100)

PART - A
(Maximum marks : 10)

Marks

I Answer the following questions in one or two sentences. Each question carries 2 marks

- 1 Define adhesive
- 2 Define step polymerisation
- 3 Define co polymers
- 4 Name the raw material used for nylon manufacture
- 5 List examples of biopolymers

(5 X 2=10)

PART - B
(Maximum marks : 30)

II Answer any five of the following questions. Each question carries 6 marks

- 1 Describe coatings
- 2 Describe suspension polymerisation
- 3 Explain calendering
- 4 Explain the manufacture of epoxies
- 5 Explain the role of additives in polymer processing
- 6 Explain melt spinning method
- 7 Describe application of plastic waste

(6 x 5 =30)

PART - C
Answer one full question from each unit
Each question carries 15 Marks

MODULE I

III

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|---|---|---|
| a | Explain melting point test for polymer analysis | 8 |
| b | Explain the manufacture of propylene | 7 |

OR

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| IV | a | Explain determination of number average molecular weight of polymers | 6 |
| | b | Explain the manufacture of caprolactum with a neat diagram | 9 |

MODULE II

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|---|---|---|---|
| V | a | Explain free radical addition polymerisation | 8 |
| | b | Describe solution polymerisation and its advantages and disadvantages | 7 |

OR

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| VI | a | Describe cationic polymerisation with an example | 7 |
| | b | Explain injection moulding | 8 |

MODULE III

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|-----|---|---|---|
| VII | a | Explain the manufacture of polyvinyl chloride with a neat diagram | 7 |
| | b | Explain the manufacture of melamine formaldehyde with a neat sketch | 8 |

OR

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| VIII | a | Explain hand lay up method for the production of reinforced plastics | 9 |
| | b | Explain the effect of reinforcement on the properties of plastics | 6 |

MODULE IV

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| IX | a | Explain dry spinning used for fibre manufacture | 7 |
| | b | Describe the manufacture of nylon 6,6 with a neat sketch | 8 |

OR

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| X | a | Explain the manufacture of polylactide(PLA) and its uses | 9 |
| | b | Explain the electro dynamic method of waste plastic separation | 6 |